Smart Budgeting and Financial Goal Management System

Use Case Specification

Project	Smart Budgeting and Financial Goal Management System — Smart Personal Budgeting and Financial Goal Management System
Prepared for	Team J
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Working draft for project implementation. Items marked "decision required" should be confirmed by the team before development.

1. Actors

Actor	Description
Visitor	A person who has not signed in; can register or sign in.
Registered user	Primary actor who manages their own finances.
System scheduler / rules engine	Internal component that calculates summaries and generates rule-based insights when data changes or a dashboard is requested.
Administrator (optional)	Technical operator for user support and configuration; not a default viewer of private transaction details.

2. Use-case overview

ID	Use case	Primary actor	Outcome
UC-01	Register account	Visitor	Account created.
UC-02	Sign in / sign out	Visitor / Registered user	Authenticated session begins or ends.
UC-03	Manage income	Registered user	Income records maintained.
UC-04	Manage expenses	Registered user	Expense records maintained.
UC-05	Manage categories	Registered user	Categories available for classifying records.
UC-06	Create and monitor budget	Registered user	Spending limits and actual usage shown.
UC-07	Manage savings goal	Registered user	Goal target and status maintained.
UC-08	Record goal contribution	Registered user	Goal progress updated.
UC-09	View dashboard	Registered user	Financial summary displayed.
UC-10	Generate/view report	Registered user	Filtered summaries and charts displayed.
UC-11	View smart insight	Registered user	Relevant rule-based tip displayed.
UC-12	Manage profile/preferences	Registered user	Permitted settings updated.

3. Use case details

UC-01 — Register account

Primary actor: Visitor

Goal: Create an account to use Smart Budgeting and Financial Goal Management System.

Main flow: 1. Visitor opens registration. 2. Enters required details. 3. System validates fields and uniqueness. 4. System securely stores account. 5. Confirmation is shown.

Alternative / exception flows: Required fields missing; identifier already used; password fails agreed policy; server error.

Postcondition: Account is created once, credentials are not stored in plain text, and user can proceed to sign in.

UC-02 — Sign in / sign out

Primary actor: Visitor / Registered user

Goal: Access personal financial data securely.

Main flow: Sign in: enter credentials; system verifies them; creates authenticated session; opens dashboard. Sign out: user selects sign out; system invalidates session.

Alternative / exception flows: Incorrect credentials; locked/expired session; network/server error.

Postcondition: Only authenticated users can access private pages; signing out ends access.

UC-03 — Manage income

Primary actor: Registered user

Goal: Record money received.

Main flow: 1. Open income form. 2. Enter amount, date, source/category and optional note. 3. Save. 4. System validates and stores record. 5. Totals refresh.

Alternative / exception flows: Invalid amount/date; missing required field; record not owned by user.

Postcondition: Record appears in list and relevant summaries update.

UC-04 — Manage expenses

Primary actor: Registered user

Goal: Record money spent.

Main flow: 1. Open expense form. 2. Enter amount, date, category and optional note. 3. Save. 4. System validates and stores record. 5. Budget and dashboard values refresh.

Alternative / exception flows: Invalid amount/date/category; record not owned by user.

Postcondition: Expense appears in list and matching budget/report calculations update.

UC-05 — Manage categories

Primary actor: Registered user

Goal: Classify income and expenses consistently.

Main flow: 1. Open categories. 2. Add, rename or delete an eligible category. 3. System validates name and type. 4. Changes are saved.

Alternative / exception flows: Duplicate name; category is in use; invalid type.

Postcondition: Categories remain correctly associated with records. Deletion behavior for categories in use must be defined.

UC-06 — Create and monitor budget

Primary actor: Registered user

Goal: Set a spending limit and track usage.

Main flow: 1. Choose category and period. 2. Enter limit. 3. Save. 4. System sums matching expenses. 5. Displays used, remaining and overspend status.

Alternative / exception flows: Invalid limit; overlapping duplicate budget; no expenses yet.

Postcondition: Budget usage matches expense records in selected category and period.

UC-07 — Manage savings goal

Primary actor: Registered user

Goal: Define a target and monitor progress.

Main flow: 1. Create goal name, target amount and optional date. 2. Save. 3. View/edit status. 4. System shows progress from contributions.

Alternative / exception flows: Invalid target; target date before creation date; goal not owned by user.

Postcondition: Goal is saved and visible only to its owner.

UC-08 — Record goal contribution

Primary actor: Registered user

Goal: Record progress towards a goal.

Main flow: 1. Open goal. 2. Enter contribution amount and date. 3. Save. 4. System updates contribution total and progress.

Alternative / exception flows: Non-positive amount; goal not owned by user; invalid date.

Postcondition: Contribution is associated with one goal and progress recalculates.

UC-09 — View dashboard

Primary actor: Registered user

Goal: Understand the current financial picture.

Main flow: 1. Sign in. 2. Open dashboard. 3. Select period if available. 4. System calculates totals, budget status and goal progress. 5. Charts/cards display.

Alternative / exception flows: No data; invalid date range; calculation service error.

Postcondition: Empty state is clear when no records exist; totals match test calculations.

UC-10 — Generate/view report

Primary actor: Registered user

Goal: Explore financial patterns.

Main flow: 1. Open reports. 2. Select date range/report type. 3. System filters owned records. 4. Charts and summaries display.

Alternative / exception flows: End date precedes start date; no matching records.

Postcondition: Report uses the selected period and only the user's records.

UC-11 — View smart insight

Primary actor: Registered user

Goal: Receive a useful observation from spending patterns.

Main flow: 1. System evaluates available records using documented rules. 2. Relevant insight is displayed with explanation and period. 3. User may dismiss it if implemented.

Alternative / exception flows: Insufficient data; no rule matches.

Postcondition: Insight is explainable, based on available data and not presented as guaranteed financial advice.

UC-12 — Manage profile/preferences

Primary actor: Registered user

Goal: Update personal settings.

Main flow: 1. Open settings. 2. Change permitted fields. 3. Save. 4. System validates and confirms.

Alternative / exception flows: Invalid value; session expired.

Postcondition: Only permitted fields change and confirmation is shown.

4. Use-case diagram (textual model)

Boundary: Smart Budgeting and Financial Goal Management System System. External actor: Visitor / Registered User. Visitor connects to Register and Sign in. Registered User connects to Manage Income, Manage Expenses, Manage Categories, Manage Budget, Manage Savings Goals, Record Goal Contribution, View Dashboard, View Reports, View Smart Insights and Manage Settings. Dashboard and reports include calculations from the user's own records. Authentication is a prerequisite for all private use cases.

5. Main end-to-end scenarios

1. Register and sign in.
2. Add income and at least two expenses in different categories.
3. Create a monthly budget for one category and verify used/remaining values.
4. Create a savings goal and record a contribution.

5. Open the dashboard and report for the same period; reconcile figures against the records.
6. Attempt to access another user's record and confirm access is denied.